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# Oxidation of commercial and surrogate bio-Diesel fuels (B30) in a jet-stirred reactor at elevated pressure: Experimental and modeling kinetic study

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**Abstract**

The oxidation of a commercial B30 (30% FAME by vol.) bio-Diesel fuel and a B30 bio-Diesel surrogate fuel (49% *n*-decane, CAS 124-18-5; 21% 1-methylnaphthalene, CAS 90-12-0; 30% methyl octanoate, CAS 111-11-5, in mole) was performed using a pressurized fused-silica jet-stirred reactor under the same initial experimental conditions (560–1030 K, 6 and 10 atm, equivalence ratios of 0.25–1.5, 10,300 ppm of carbon). The results of this series of experiments consisted of concentration profiles of reactants, stable intermediates and products measured as a function of temperature by low-pressure probe sampling followed by Fourier transform infrared absorption spectrometry and gas chromatography analyses. The results obtained with the commercial and surrogate B30 mixtures were compared with each other, showing that the mixture *n*-decane/1-methylnaphthalene/methyl octanoate 49/21/30% in mole is an excellent simple B30 Diesel fuel surrogate. A detailed chemical kinetic reaction mechanism consisting of 7748 reactions involving 1964 species was proposed based on previous chemical kinetic reaction mechanisms for the oxidation of *n*-decane, methyl octanoate, and 1-methylnaphthalene under similar conditions. The kinetic modeling showed reasonable agreement between the present data and computations over the entire range of conditions considered in this study.
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*Keywords:* Bio-Diesel; *n*-Decane; 1-Methylnaphthalene; Methyl octanoate; Kinetic modeling

## 1. Introduction

It is recognized that among other automotive engines, Diesel engines contribute significantly to overall carbon dioxide emissions whereas concerns about green-house effect and air pollution favor the investigation of sustainable and environment-friendly Diesel fuels. Biofuels such as fatty acid methyl esters (FAME) are mixed in variable quantities (e.g. B5 contains 5% in volume of FAME) with fossil Diesel fuel [1–3]. Reduction of engines emissions in terms of carbon oxides and polyaromatic hydrocarbons (PAH) have been reported, indicating bio-Diesel may help preserving our environment [3–5]. The so-called bio-Diesel is a mixture of FAME produced from transesterification of

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triglycerides (oils) with methanol. Current bio-Diesel fuels are mixtures of ca. $C_{12}$–$C_{22}$ highly saturated carbon-chain esters. Their complex composition implies the use of surrogate model-fuels for simulating their combustion kinetics. Whereas early kinetic studies have demonstrated a strong similitude between the oxidation of rapeseed oil methyl esters (RME) and that of *n*-hexadecane, long-chain methyl esters exhibiting cool-flames were also proposed as bio-Diesel model-fuels [6–8]. A fossil Diesel fuel consists of an even more complex mixture of thousands of medium–high molecular weight hydrocarbons [9] that participate in thousands of pyrolysis and oxidation reactions. Therefore, surrogates [9,10] are also employed to represent Diesel fuel with a limited number of components [10]. In Europe, a surrogate Diesel fuel called the 'IDEA' fuel, consisting of 70% *n*-decane and 30% 1-methyl naphthalene was formulated previously as part of the Integrated Development on Engine Action (IDEA) program [11]. This fuel mixture matches both the physicochemical properties and combustion behavior of a conventional Diesel fuel. The IDEA fuel has properties similar to those of a conventional Diesel fuel, i.e. a normal density of 0.798 g/L at 20 °C, a CN of ca. 53, and a hydrogen-to-carbon ratio of 1.8.

In this study, the kinetics of oxidation of a commercial B30 bio-Diesel fuel and a B30 surrogate bio-Diesel fuel were measured and compared. These experiments were performed in a jet-stirred reactor (JSR), in order to: (1) provide new information on the kinetic of oxidation of a B30 bio-Diesel fuel over a wide range of conditions, (2) verify the chemical kinetics of oxidation of a simple B30 surrogate can represent that of a commercial B30 Diesel fuel, and (3) propose and validate a detailed kinetic reaction mechanism for the oxidation of a B30 bio-Diesel fuel from low to high temperatures.

## 2. Experimental

We used a jet-stirred reactor (JSR) setup similar to that employed earlier [10,12,13]. It consisted of a fused-silica (to minimize wall catalytic reactions) sphere of 33 $cm^3$ in volume, equipped with four nozzles of 1 mm i.d. for the injection of the gases achieving the stirring. A nitrogen flow of 100 L/h was used to dilute the fuel and avoid its pyrolysis before admission into the reactor. All the gases were preheated before injection to minimize temperature gradients inside the JSR. The liquid fuels were atomized and vaporized before injection into the reactor. The fuel and oxygen were diluted by a flow of nitrogen (<50 ppm of $O_2$; <1000 ppm of Ar; <5 ppm of $H_2$), and mixed at the entrance of the injectors. High purity hydrocarbons from Aldrich were used: *n*-decane (>99% pure), 1-methyl naphthalene (>95% pure), and methyl octanoate (99% pure). The experiments were performed at steady state, at a constant mean residence time, the reactants flowing continuously in the reactor. The temperature of the gases inside the JSR ranged from 560 to 1030 K. A high degree of dilution was used (10,300 ppm of carbon), minimizing temperature gradients in the JSR and heat release. Thermocouple (0.1 mm Pt/Pt–Rh 10% wires inside a thin-wall silica tube) measurements showed a good thermal homogeneity along the vertical axis of the JSR (gradient <3 K/cm). The reacting mixtures were sampled via a low-pressure fused-silica sonic probe. On-line Fourier transform infrared (FTIR) analyses of the reacting gases were performed by connecting the sampling probe to a temperature-controlled gas cell (140 °C, 2 m path length, 0.5 bar, and a 1 $cm^{-1}$ resolution) via a stainless steel heated line. Gas chromatographs (GC) equipped with capillary columns (DB-5 ms, DB-624, Plot $Al_2O_3$/KCl, Carboplot-P7), mass spectrometry (MS), thermal conductivity detector (TCD), and flame ionization detector (FID), were used for species measurements. Compounds identification was made via GC/MS analyses (Saturn 2000 and V1200, Varian) using electron impact ionization mode (70 eV). The samples were analyzed via on-line GC–MS/FID (for condensable products) and off-line by GC–MS/FID/TCD (for permanent gases and high vapor-pressure species) after collection and storage in 1 L Pyrex bulbs. A good repeatability of the measurements and a reasonably good carbon balance (100 ± 10%) were obtained in this series of experiments.

## 3. Modeling

The computations were performed using the PSR computer code [14]. The detailed chemical kinetic reaction mechanism used here is based on previous studies of the oxidation of methyl octanoate [15], large alkanes [9,10,16], and 1-methylnaphtalene [13]. Although cross-reactions describing the interactions between the main fuel components are of secondary importance, they were considered previously [17] for modeling the oxidation of commercial and surrogate Diesel fuels under conditions similar to those of the present study. They included metathesis reactions of *n*-decane with phenyl, benzyl, 1-naphtylmethyl, 1-naphtyl, and indenyl radicals. The reactions of 1-vinylnaphtalene with $HO_2$ and of decyl radicals with 1-naphtaldehyde were also considered [17]. To be consistent with this previous study, reactions of $C_{10}H_{21}O_2$ with toluene, 1-methylnaphtalene, 1-naphtylmethyl, 1-naphtaldehyde, benzyl, phenyl, and 1-naphtyl were added here. The kinetics of these added reactions were assumed based on similar reactions used in the kinetic base-set [18] and in the literature [19]. The proposed kinetic reaction mechanism consisting of 7748 reversible reactions

involving 1964 species represents the first attempt to propose a kinetic scheme for the oxidation of Diesel–bio-Diesel fuel mixtures. It is available from the authors and as Supplementary data. The rate constants for the reverse reactions were computed from the forward rate constants and the equilibrium constants calculated using the appropriate thermochemical data [10,13,20,21].

## 4. Results and discussion

### 4.1. Experiment results

The oxidation of a B30 bio-Diesel fuel surrogate, 49% *n*-decane, 21% 1-methyl naphthalene, and 30% methyl octanoate in mole, i.e. $C_{10.3}H_{18.4}O_2$, and a commercial low-sulfur B30 bio-Diesel fuel (CN 54.8, 84.1% C, 12.9% H, and 3% O by wt., $C_{16.47}H_{30.83}O_{0.5}$, $S < 6$ ppmw, $d = 845$ g/L at 15 °C) were studied in a jet-stirred reactor over a wide range of conditions (Table 1). The FAME fraction of the commercial B30 was rapeseed oil methyl ester. In these experiments, the temperature was varied step-wise in the range 560–1030 K, keeping the mean residence time constant and equal to 0.6 s at 6 atm, and to 1 s at 10 atm. This allowed the observation of the cool-flame oxidation regime, the negative temperature coefficient (NTC) regime, and the high temperature oxidation regime. The experiments were performed for several equivalence ratios, from fuel-lean to fuel-rich conditions (i.e. $\varphi = 0.25$–1.5). More than 20 species were identified and measured by FTIR, CG–MS/FID/TCD. Experimental concentration profiles were obtained for $H_2$, $H_2O$, $O_2$, CO, $CO_2$, $CH_2O$, $CH_4$, $C_2H_6$, $C_2H_4$, $C_2H_2$, methanal, ethanal, $C_3H_6$, 1-$C_4H_8$, 1,3-$C_4H_6$, 1-$C_5H_{10}$, 1-$C_6H_{12}$, 1-$C_8H_{16}$, *n*-decane, methyl octanoate, and 1-methylnaphthalene. Other minor species detected at ppm levels were not quantified nor used in the modeling.

The concentration profiles measured from the oxidation of the commercial B30 and the B30 surrogate over the low-, intermediate-, and high temperature oxidation regimes were compared. As can be seen from Fig. 1, the concentration profiles obtained for CO, $CO_2$, $H_2O$, and $O_2$ during the oxidation of the two fuels are very similar over the entire range of experimental conditions. Similar agreement was observed at the other equivalence ratios. Also it can be noted from this figure that the commercial Diesel fuel used in the B30 mixture reacts similarly. The concentrations of $C_2H_4$ and $C_3H_6$ produced by the oxidation of the B30 surrogate fuel are slightly higher than those obtained with the B30 fuel. This trend can be rationalized by considering the higher fraction of *n*-alkanes in the surrogate yielding more ethylene and propene via β-scission. Nevertheless, these results indicate the present B30 surrogate is appropriate for studying the kinetics of oxidation of a commercial B30 fuel.

The present results also showed that increasing the initial concentration of oxygen, keeping the initial fuel mole fraction constant, resulted in an increase of the rate of oxidation of the fuel, in line with previous results obtained in similar conditions with a variety of fuels, including a synthetic Diesel fuel [10]. It was also observed that the importance of the NTC increased when the initial concentration of oxygen was decreased. Moreover, it was noticeable that decreasing the initial oxygen concentration yielded higher maximum concentrations of stable intermediates at high temperature (>800 K). For example, at 10 atm, methane maximum mole fraction reached 550 ppm at an equivalence ratio of 1.5 whereas it was only 120 ppm at $\varphi = 0.25$. In the same conditions, ethylene maximum mole fraction varied from 270 ppm at $\varphi = 0.25$ to 910 ppm at $\varphi = 1.5$. Also, propene maximum mole fraction varied from 54 ppm at $\varphi = 0.25$ to 140 ppm at $\varphi = 1.5$. Similarly, 1-butene maximum mole fraction varied from 19 ppm at $\varphi = 0.25$ to 47 ppm at $\varphi = 1.5$. These variations resulted from the reduction of the radical pool with decreasing oxygen initial concentration. They are also in line with observations made previously for the oxidation of a variety of hydrocarbon fuels, including commercial fuels [9,10,17] in similar conditions.

The effect of total pressure on the fuel oxidation was studied by comparing the present results obtained at 6 and 10 atm. They indicated that the

Table 1
Experimental conditions in the JSR.

| Initial concentrations (in ppm for the fuel, in mol fraction for $O_2$ and $N_2$) | | | | | $\varphi$ | $P$ (atm) |
|---|---|---|---|---|---|---|
| B30 | *n*-$C_{10}H_{22}$ | $C_{11}H_{10}$ | $C_9H_{18}O_2$ | $O_2$ | | |
| 600 | – | – | – | 0.0574 | 0.25 | 10 |
| 600 | – | – | – | 0.0287 | 0.5 | 10 |
| 600 | – | – | – | 0.0144 | 1 | 10 |
| 600 | – | – | – | 0.0096 | 1.5 | 10 |
| – | 490 | 210 | 300 | 0.0597 | 0.25 | 10 |
| – | 490 | 210 | 300 | 0.0284 | 0.5 | 6, 10 |
| – | 490 | 210 | 300 | 0.0142 | 1 | 6, 10 |
| – | 490 | 210 | 300 | 0.0095 | 1.5 | 10 |

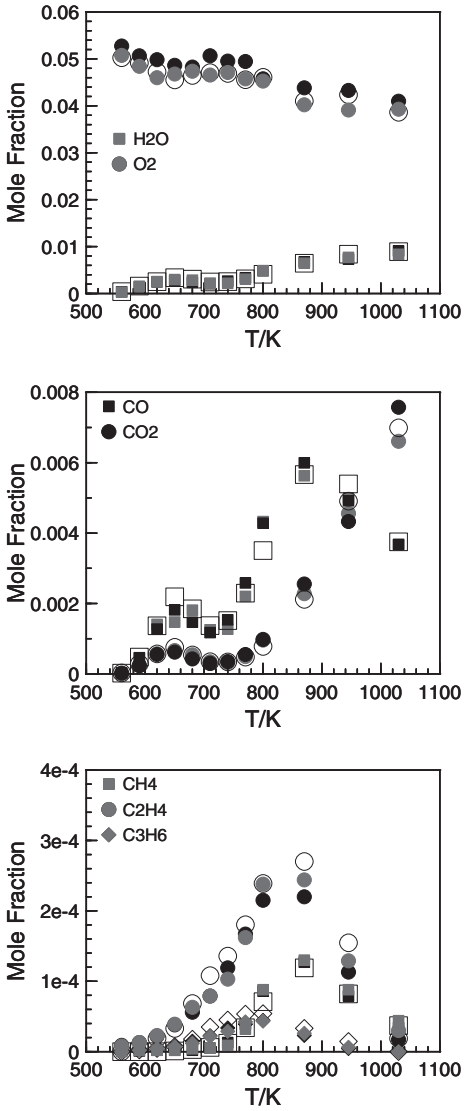


Fig. 1. Experimental species concentration profiles from the oxidation of a commercial Diesel fuel (filled symbols), the B30 fuel (grey symbols), the B30 surrogate (empty symbols) in a JSR at 10 atm, $\varphi = 0.25$, and $\tau = 1$s.

importance of the NTC is reduced when the total pressure increases. Once more, such a trend was observed previously for the oxidation of *n*-heptane in a JSR under similar conditions [22].

### 4.2. Modeling results

The concentration profiles obtained for the oxidation of the B30 surrogate fuel were compared to the model predictions. It should be noted that the present model was also successfully tested for the oxidation of pure *n*-decane, pure methyl

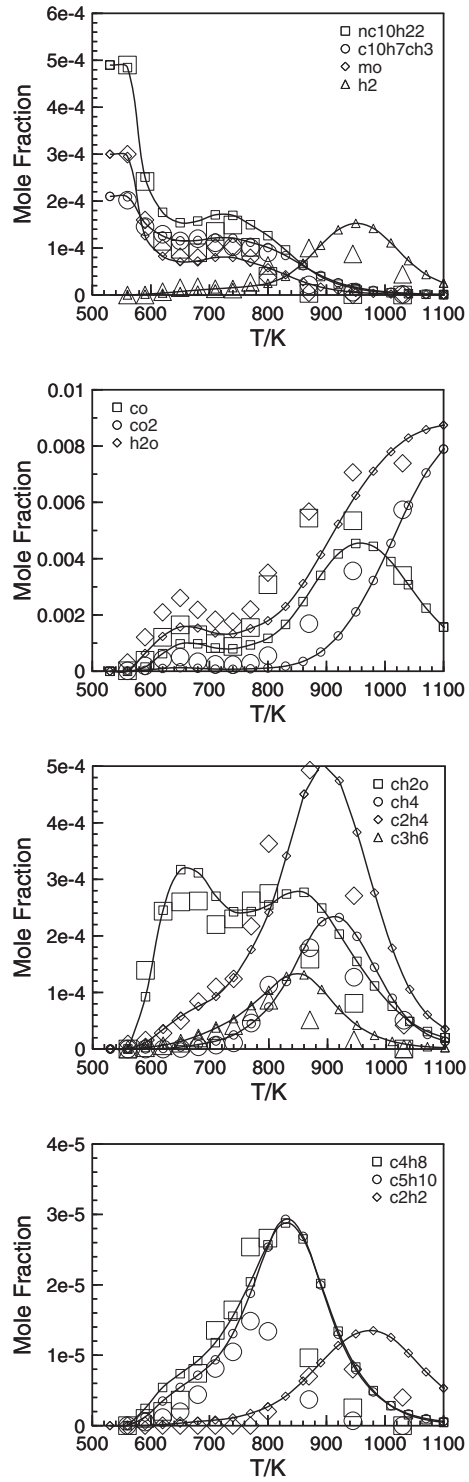


Fig. 2. The B30 surrogate Diesel fuel oxidation in a JSR at 10 atm, $\tau = 1$s, and $\varphi = 0.5$. The experimental data (large symbols) are compared to the computations (lines with small symbols).

octanoate, and pure 1-methylnaphtalene under similar JSR conditions [13,16,23,24]. Furthermore, the proposed model, not including the methyl octanoate chemistry, was used to simulate the oxidation of commercial and surrogate Diesel fuels [17] Figs. 2–6 show examples of the comparison between the present experimental and modeling results. According to the computations, under the present conditions, in the cool-flame regime (ca. 560–720 K), both methyl octanoate, 1-methyl naphthalene, and *n*-decane are oxidized resulting in the formation of many products. As can be seen from these figures, large amounts of CO, $CO_2$, $H_2O$, and formaldehyde are formed under cool-flame conditions (peak concentration observed at ca. 650 K) and high temperature (peak

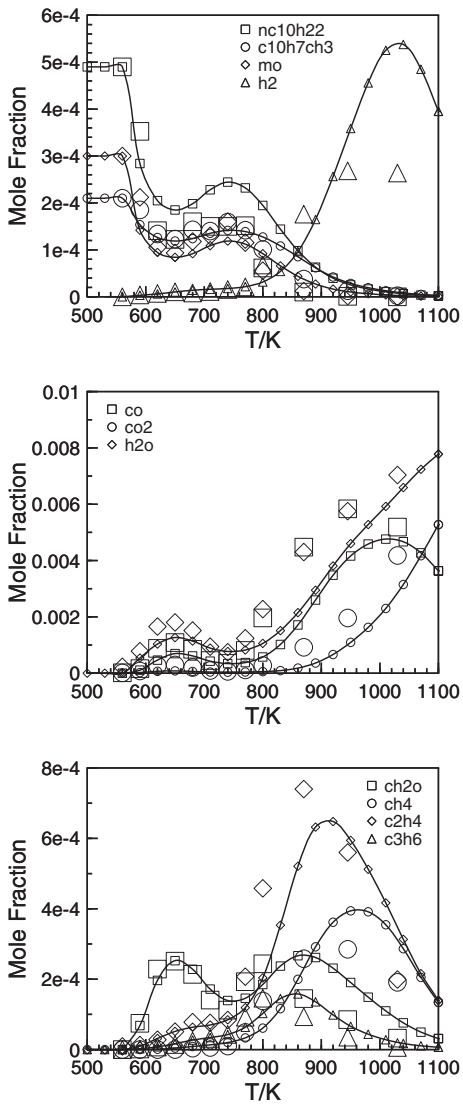


Fig. 3. The B30 surrogate Diesel fuel oxidation in a JSR at 10 atm, $\tau = 1$s, and $\varphi = 1$.

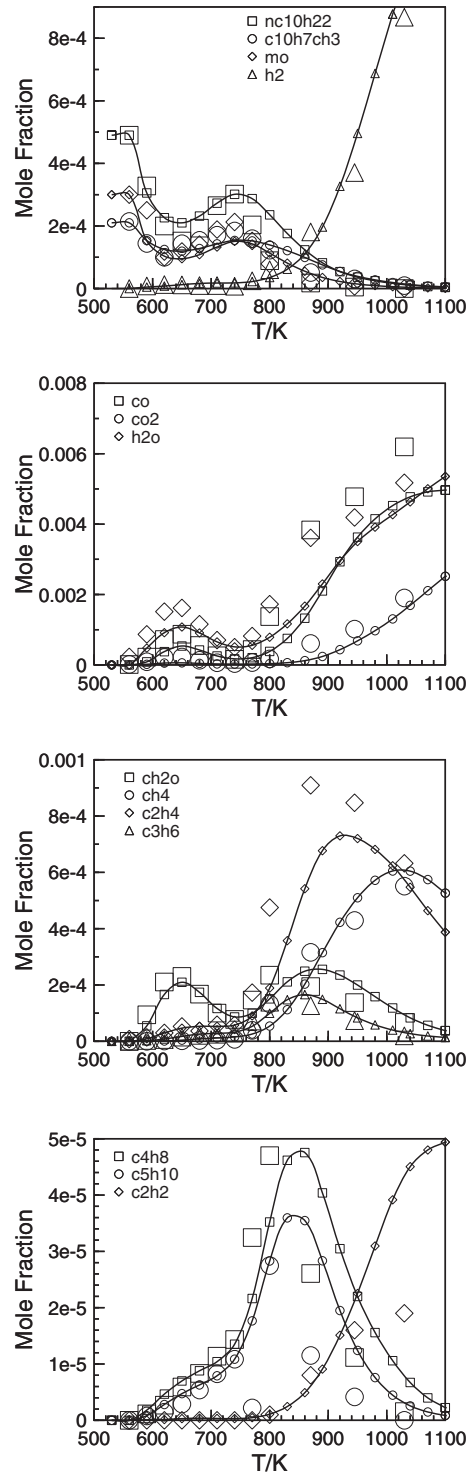


Fig. 4. The B30 surrogate Diesel fuel oxidation in a JSR at 10 atm, $\tau = 1$s, and $\varphi = 1.5$.

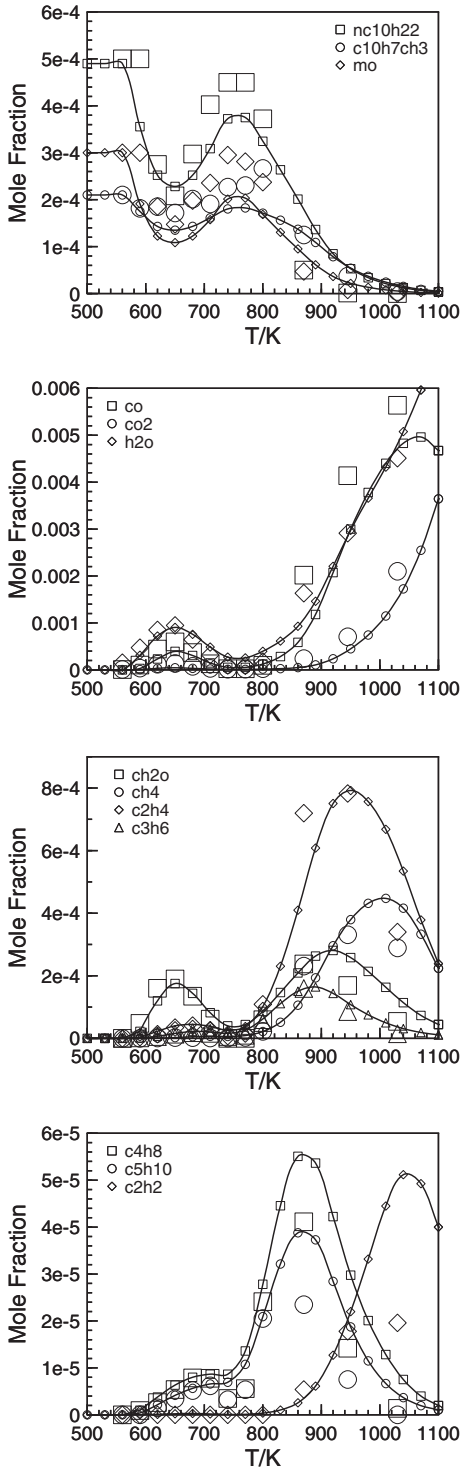


Fig. 5. The B30 surrogate Diesel fuel oxidation in a JSR at 6 atm, $\tau = 0.6$s, and $\varphi = 1$.

concentration of CO and $CH_2O$ observed at ca. 850–900 K). The kinetic model reasonably well

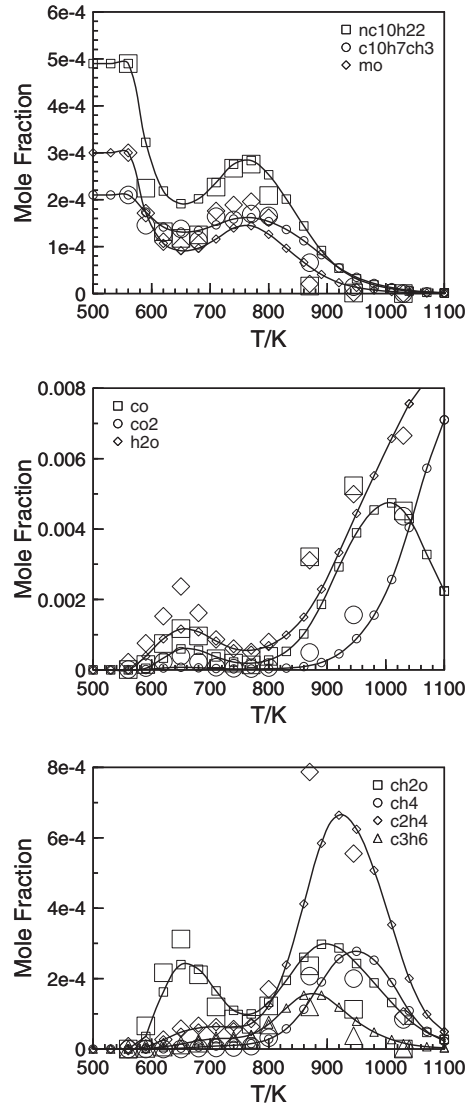


Fig. 6. The B30 surrogate Diesel fuel oxidation in a JSR at 6 atm, $\tau = 0.6$s, and $\varphi = 0.5$.

predicts these results in most of the cases although it seems to overestimate $CH_2O$ mole fraction in fuel-lean conditions (Fig. 2). The consumption of the fuel components showing a cool-flame oxidation regime (ca. 560–800 K), a negative temperature behavior (ca. 650–800 K), and a transition to high temperature oxidation (>800 K) is also in agreement with the modeling. This is an interesting result since under similar conditions, in absence of other hydrocarbons or methyl ester, 1-methyl naphthalene would not react. Here, its oxidation occurs thanks to the radical pool provided through the oxidation of *n*-decane and methyl octanoate.

According to the present computations, at 620 K and in fuel-lean conditions ($\varphi = 0.5$ at 10 atm), OH radicals are mostly responsible for the oxidation of *n*-decane (ca. 95%), methyl octanoate (ca. 89%), and 1-methylnaphtalene (ca. 80%) via $n\text{-}C_{10}H_{22} + OH = C_{10}H_{21} + H_2O$, $C_9H_{18}\text{-}O_2 + OH = C_9H_{17}O_2 + H_2O$, and $C_{10}H_7CH_3 + OH = C_{10}H_7CH_2 + H_2O$. Under these conditions, their formation mainly occurs via the decomposition of alkylhydroperoxy ($O_2QOOH$ and $OQ'\text{-}OOH$) deriving from the oxidation of *n*-decane and methyl octanoate. Under these conditions, where oxidation processes of alkyl radicals are more important than β-scissions, little ethylene or propene formation is observed. Actually, the modeling indicates ethylene is mainly produced by decomposition of $C_2H_4OOH$.

Above ca. 750 K, the transition to the high temperature oxidation regime occurs. There, the fuel is rapidly consumed through metathesis reactions with OH and larger amounts of products are formed. Again, the model predicts the experimentally observed overall reactivity of the fuel and products' formation, although it tends to underestimate the overall rate of oxidation above ca. 800 K. This behavior results from the too strong inhibiting effect of 1-methylnaphtalene on *n*-decane and methyl octanoate oxidation that could be due to missing cross-reactions in the model. We did not attempt to improve the present simulations by modifying the kinetic parameters used in previous modeling efforts in order to keep this model valid for representing the neat oxidation of the surrogate fuel components, i.e. *n*-decane, 1-methylnaphtalene, and methyl octanoate. The computations indicated that the computed results were moderately influenced by cross-reactions, in agreement with previous findings for the oxidation of Diesel and surrogate Diesel fuels under similar conditions [17].

The present kinetic modeling indicated that at 1040 K, OH radicals are still mostly responsible for the oxidation of *n*-decane (ca. 80%), methyl octanoate (ca. 80%), and 1-methylnaphtalene (ca. 89%) via the same reactions as above. The reactions of *n*-decane with O (ca. 10%) and H (ca. 5%) also contribute to its consumption. Also, methyl octanoate reacts with H (ca. 8%). Similarly, H-atoms also consume 1-methylnaphtalene (ca. 8%). Under these conditions, the production of ethylene mainly occurs via β-scissions of alkyl radicals (1-butyl and 1-propyl 30%) whereas the oxidation of ethyl radicals by $O_2$ also contributes to ethylene formation (20%).

Local, first-order sensitivity analyses were also performed. They showed that at 620 K, besides the $C_0$–$C_2$ reactions, the overall reactivity is positively sensitive to the rates of oxidation of *n*-decane by OH, and the peroxidation of methyl octanoate radicals, i.e. to reactions $CH_3(CH_2)_5CH(\bullet)C(=O)OCH_3 + O_2 \Longleftrightarrow CH_3(CH_2)_5CH(OO\bullet)C(=O)OCH_3$ and $CH_3(CH_2)_5CH(OO\bullet)C(=O)OCH_3 \Longleftrightarrow CH_3\text{-}CH_2CH_2CH_2CH(\bullet)CH_2CH(OOH)C(=O)OCH_3$. As expected, at 1040 K, the system is mostly sensitive to the kinetics of the $C_0$–$C_1$ sub-scheme, i.e. $H + O_2 \Longleftrightarrow OH + O$, $HO_2 + OH \Longleftrightarrow H_2O + O_2$, and $CO + OH \Longleftrightarrow CO_2 + H$.

## 5. Conclusion

The three main goals of this study were achieved. (i) The kinetics of oxidation of a commercial B30 fuel and a B30 surrogate fuel (49% *n*-decane, 30% methyl octanoate, 21% 1-methylnaphthalene in mole) were studied in dilute conditions (10300 ppm of C) using a pressurized fused-silica jet-stirred reactor operating over a broad range of conditions (560–1030 K, 6 and 10 atm, equivalence ratios of 0.25–1.5). (ii) The present experimental results consisting of concentration profiles of reactants, stable intermediates and products obtained at fixed mean residence time as a function of temperature showed the two fuels oxidize very similarly, allowing the use of the three-component surrogate, i.e. *n*-decane-methyl octanoate-1-methylnaphthalene, for kinetic studies. (iii) A chemical kinetic modeling of the present experiments was performed using a detailed kinetic scheme (7748 reactions involving 1964 species) and the B30 surrogate was used as model-fuel. The kinetic modeling results were found in reasonable agreement between the present JSR data. The main reaction paths were delineated showing the importance of OH radicals' production and consumption. The modeling showed little interaction between the fuel components at high temperature whereas the cool-flame oxidation of 1-methylnaphtalene occurs via the radical pool generated via the oxidation of the two other more reactive fuel components, i.e. *n*-decane and methyl octanoate. Further kinetic modeling of B30 bio-Diesel ignition delays and flame speeds are still needed to fully assess the validity of the proposed model.

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## Appendix A. Supplementary data

This publication includes the following supplemental material: the proposed chemical kinetic reaction mechanism in CHEMKIN format and the associated thermochemical data in CHEMKIN format. Supplementary data associated with

this article can be found, in the online version, at doi:10.1016/j.proci.2010.05.096.

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